ANNIHILATION OF EXT ONE BY AN IDEAL IS EQUIVALENT TO THE MULTIPLICATION MAPS ON THE FIRST TERM FACTORING THROUGH THE DIFFERENTIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted proof span begins with a blank line, which is reproduced. The author writes the incomplete sentence 'Since is annihilated by a in I we see that we get the dotted arrow'; it is reproduced unchanged rather than silently repaired. The four references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The description of maps out of a two-term complex in the derived category. invoked at proof line 22483. Lets an element of Ext^1_R(K, N) be represented by a map from the first term, which the computation a varphi = varphi o h o d_K^{-1} relies on.
- The representation lemma for a complex concentrated in two degrees, with a projective or finite free first term. invoked at proof lines 22495 and 22506. Supplies the normal form of the representing complex in conditions (2) and (5).
- Ext^1_R(K, N) as classes of maps and the identification of the class of the identity on K^{-1}. used at proof lines 22484-22486. Is the choice that turns the annihilation hypothesis into the existence of the dotted arrow.

Source reference. Original lemma 22338: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22338–22338. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 22304: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22304–22304. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let R be a ring and let $I \subset R$ be an ideal. Let $K \in D(R)$. Assume $H^i(K) = 0$ for $i \notin \{-1, 0\}$. The following are equivalent*

1. $\mathrm{Ext}^1_R(K, N)$ *is annihilated by I for all R-modules N,*
2. *K can be represented by a complex $K^{-1} \to K^0$ with K^0 free such that for any $a \in I$ the map $a : K^{-1} \to K^{-1}$ factors through $d_K^{-1} : K^{-1} \to K^0$,*
3. *whenever K is represented by a two term complex $K^{-1} \to K^0$ with K^0 projective, then for any $a \in I$ the map $a : K^{-1} \to K^{-1}$ factors through $d_K^{-1} : K^{-1} \to K^0$.*

If R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$, then these are also equivalent to

4. $\mathrm{Ext}^1_R(K, N)$ *is annihilated by I for every finite R-module N,*
5. *K can be represented by a complex $K^{-1} \to K^0$ with K^0 finite free and K^{-1} finite such that for any $a \in I$ the map $a : K^{-1} \to K^{-1}$ factors through $d_K^{-1} : K^{-1} \to K^0$.*

Proof. Assume (1) and let $K^{-1} \to K^0$ be a two term complex representing K with K^0 projective. We will use the description of maps in $D(R)$ out of $K^\bullet$ given in Lemma 22338 without further mention. Choosing $N = K^{-1}$ consider the element ξ of $\mathrm{Ext}^1_R(K, N)$ given by $\mathrm{id}_{K^{-1}} : K^{-1} \to K^{-1}$. Since is annihilated by $a \in I$ we see that we get the dotted arrow fitting into the following commutative diagram

$$\begin{array}{ccc} K^{-1} & \xrightarrow{d_K^{-1}} & K^0 \\ {\scriptstyle a}\downarrow & \swarrow_{h} & \\ K^{-1} & & \end{array}$$

This proves that (3) holds. Part (3) implies (2) in view of Lemma 22304 part (1). Assume $K^\bullet$ is as in (2) and N is an arbitrary R-module. Any element ξ of $\mathrm{Ext}^1_R(K, N)$ is given as the class of a map $\varphi : K^{-1} \to N$. Then for $a \in I$ by assumption we may choose a map h as in the diagram above and we see that $a\varphi = \varphi \circ a = \varphi \circ h \circ \mathrm{d}_K^{-1}$ which proves that $a\xi$ is zero in $\mathrm{Ext}^1_R(K, N)$. Thus (1), (2), and (3) are equivalent.

Assume R is Noetherian and $H^i(K)$ is a finite R-module for $i = -1, 0$. Part (3) implies (5) in view of Lemma 22304 part (2). It is clear that (5) implies (2). Trivially (1) implies (4). Thus to finish the proof it suffices to show that (4) implies any of the other conditions. Let $K^{-1} \to K^0$ be a complex representing K with K^0 finite free and K^{-1} finite as in Lemma 22304 part (2). The argument given in the proof of (2) $\Rightarrow$ (1) shows that if $\mathrm{Ext}^1_R(K, K^{-1})$ is annihilated by I, then (1) holds. In this way we see that (4) implies (1) and the proof is complete. $\square$